# Rainfall Monitoring System

## Flutter Mobile Application — Design & Architecture Specification

| | |
|---|---|
| **Client** | Govt. of Chhattisgarh — Water Resources Dept. |
| **Scope** | Mobile-only · Field Operator · Block / District / State Officers |
| **Platform** | Flutter (Android + iOS) · Tablet responsive · Dark mode |
| **MVP Geography** | Raipur District · 3 Blocks · 8–10 Villages · 10 Stations |
| **Production Target** | Chhattisgarh · 33 Districts · 734 Stations · Thousands of users |
| **Document** | v1.0 — Source-of-truth design spec |

# 1. Requirements Traceability

Every requirement extracted from the MVP Design Document is mapped to a screen, component, and architectural layer. Nothing is dropped.

| # | Requirement (PDF) | Realised by | Layer |
|---|---|---|---|
| R1 | Manual rainfall data entry | Rainfall Entry screen (S07) | Form widgets · Riverpod RainfallController |
| R2 | GPS-based station verification (≤20 m) | GPS Verification screen (S06) | Geolocator + distanceBetween() |
| R3 | Role-based hierarchical access | Auth flow + RoleGuard router middleware | GoRouter redirect + Riverpod AuthState |
| R4 | Offline data storage | Offline Queue (S09), local Hive box | HiveAdapter + SyncWorker |
| R5 | GIS visualization (state, district, block) | GIS dashboards (S10, S11), block map (S12) | flutter_map + MarkerClusterLayer |
| R6 | Missing station monitoring | Block/District dashboards · Missing list | Aggregation provider · status=red filter |
| R7 | Dashboard aggregation Station→Village→Block→District→State | All officer dashboards | AggregationService + Riverpod family providers |
| R8 | Auth: Mobile+OTP and Username+Password | Login (S02), OTP (S03) | AuthRepository (mock) → AuthApi (prod) |
| R9 | Device binding (simulated in MVP) | Login + Profile screens | DeviceInfoService + secure storage |
| R10 | Fields: rainfall, timestamp, lat, lng, stationId, remarks | Rainfall Entry form (S07) | RainfallEntry model + Hive type 0 |
| R11 | Sync states Synced / Pending / Failed | Offline queue chips, sync center | SyncStatus enum |
| R12 | Station status Green / Yellow / Red | Map markers, station cards | StationStatus enum + theme tokens |
| R13 | State map with district color coding | State GIS (S11) choropleth | GeoJSON layer · stop colors by compliance |
| R14 | District map with station markers | District GIS (S10) | Marker layer + bottom sheet |
| R15 | District metrics (total, reported, missing, pending, avg, max) | District/Block dashboards KPI strip | KpiCard widget · MetricsProvider |
| R16 | Data flows upward Station→…→State | AggregationService | Background isolate aggregation |
| R17 | Scalable to 734 stations / all CG | Map clustering, pagination, lazy lists | MarkerClusterLayer · sliver pagination |
| R18 | Flutter, Riverpod, GoRouter, flutter_map, Geolocator, Hive, fl_chart, permission_handler | See §10 Architecture | pubspec.yaml pinned versions |
| R19 | Profile / Settings / Help | Profile screen (S14) | SettingsRepository |

| # | Requirement (PDF) | Realised by | Layer |
|---|---|---|---|
| R20 | Future: push, alerts, heatmaps, audit, real-time | Architecture leaves seams: NotificationService, AuditLogger, RealtimeChannel | Documented in §13 Scalability |

# 2. User Personas & Journeys

## 2.1 Personas

| Persona | Context | Primary Goal | Constraints |
|---|---|---|---|
| Ramesh — Field Operator | Village station, low-end Android, weak network, outdoors | Submit rainfall daily before 08:30 | One station only; needs offline, GPS, large tap targets |
| Sunita — Block Officer | Block HQ, mid-range phone, intermittent 4G | Ensure 100% reporting across her block | No edit rights; needs missing-station alerts |
| Vikram — District Officer | Raipur collectorate, tablet + phone | Monitor blocks, intervene on gaps | Drill-down from district → block → station |
| Anjali — State Admin | Raipur HQ, tablet, dual-SIM | Statewide compliance, escalations | Needs choropleth, trend analytics, exports |

## 2.2 Field Operator Journey (happy path)

Splash → Login (Mobile+OTP) → OTP Verify → Home Dashboard → My Station → GPS Verification (≤20 m) → Rainfall Entry → Success → Marker turns Green on GIS.

## 2.3 Field Operator Journey (offline)

Login (cached) → Home → My Station → GPS Verification (works offline) → Entry → Saved locally as **Pending Sync** (yellow) → On reconnect, SyncWorker uploads → status becomes Synced (green).

## 2.4 Block / District / State Officer Journey

Login → Role-aware Dashboard → Filter (date, block, village) → GIS map → Tap missing station → Bottom sheet → Tap Contact Operator / Drill down.

# 3. Information Architecture

A single Flutter app serves all four roles. The navigation tree below is enforced by GoRouter **redirect** based on the authenticated role; unreachable branches are pruned at build time per role to keep bottom-nav focused.

```
ROOT ■■■ /splash ■■■ /auth ■ ■■■ /login (Mobile+OTP | Username+Password tabs) ■ ■■■ /otp ■■■
/operator (Field Operator shell — bottom nav) ■ ■■■ /home ■ ■■■ /stations (assigned station
list — single in MVP, scalable) ■ ■ ■■■ /:id ■ ■ ■■■ /verify (GPS verification) ■ ■ ■■■
/entry (rainfall submission) ■ ■■■ /queue (offline queue + sync center) ■ ■■■ /history ■ ■■■
/profile ■■■ /block (Block Officer shell — bottom nav) ■ ■■■ /dashboard ■ ■■■ /villages/:id ■
■■■ /stations/:id ■ ■■■ /map (GIS — block scope) ■ ■■■ /reports ■ ■■■ /profile ■■■ /district
(District Officer shell) ■ ■■■ /dashboard ■ ■■■ /blocks/:id ■ ■■■ /stations/:id ■ ■■■ /map
(GIS — district scope) ■ ■■■ /analytics ■ ■■■ /reports ■ ■■■ /profile ■■■ /state (State Admin
shell) ■ ■■■ /dashboard ■ ■■■ /districts/:id ■ ■■■ /map (GIS — state choropleth) ■ ■■■
/compliance ■ ■■■ /analytics ■ ■■■ /reports ■ ■■■ /profile ■■■ /shared ■■■ /notifications
■■■ /settings ■■■ /help
```

# 4. Mobile Navigation Flow

The diagram below shows the canonical state machine. Solid arrows are user-driven; dashed arrows are system-driven (auth, sync, GPS).

```
[Splash] ■■auth?■■■ [Login] ■■submit■■■ [OTP] ■■verify■■■ [RoleRouter] ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ ▼
▼ ▼ ▼ ▼ [Operator/Home] [Block/Dashboard] [District/Dash] [State/Dash] [Shared/Notif] ■ ■ ■ ■
■■■[Station Detail]■■■[GPS Verify]■■[Entry]■■[Success]■ ■ ■ ■■■[Offline
Queue]■■■■■■■■■■■■■■■■■■■■■■■ pending ■■■■■ ■ ■■■[History] [Profile] [Settings]
```

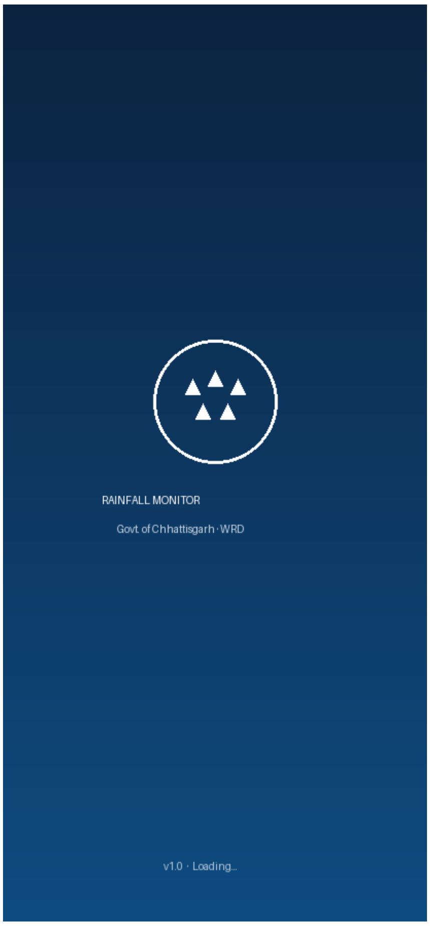
## 4.1 Deep links

- **rms://station/RP004** — opens station detail (operator) or station drill-down (officer).
- **rms://gis?d=raipur&b;=abhanpur&date;=2026-06-10** — GIS pre-filtered.
- **rms://alert/missing/RP008** — from push notification, opens action sheet.

# 5. Screen-by-Screen UI Design

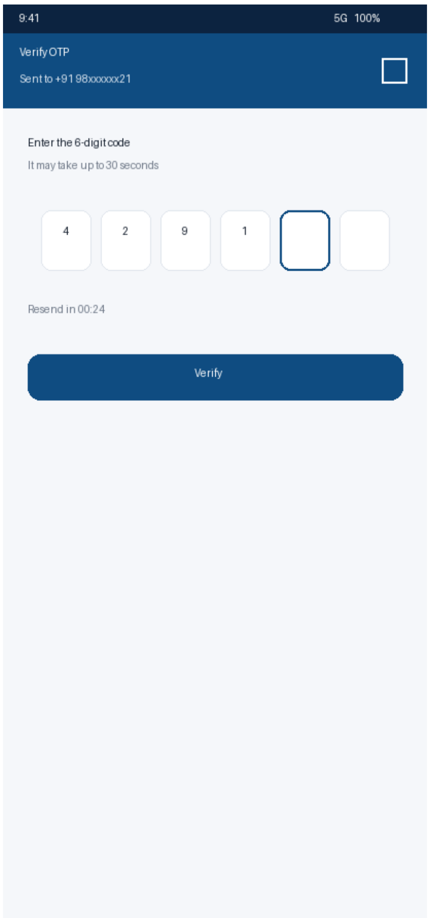
Every screen specified in the MVP document is included. Each card lists purpose, key widgets, data dependencies, and edge states.

## 5.1 Splash · Login · OTP




S01 — Splash

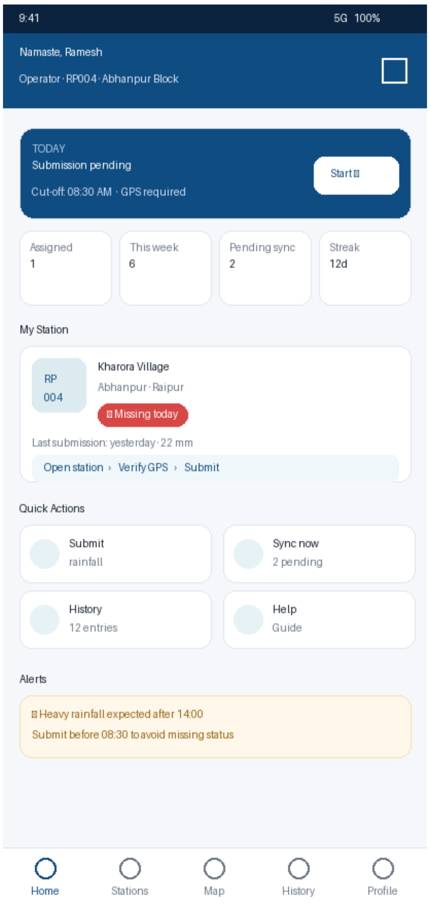

S02 — Login (Mobile+OTP / Username)



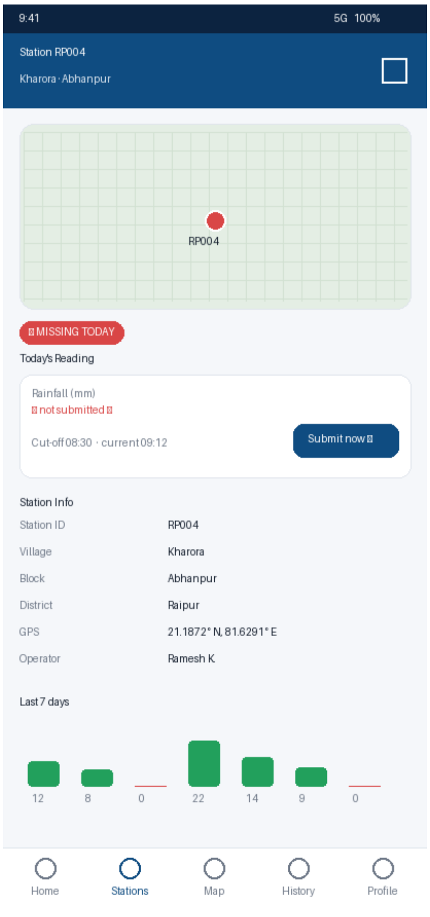

S03 — OTP Verification

| Screen | Key widgets | Data / State | Edge cases |
|---|---|---|---|
| S01 Splash | RmsLogo, version, progress | BootstrapController checks token, role, device | Token expired → /login; no internet → cached role still routes |
| S02 Login | SegmentedTabs, RmsTextField, RmsPasswordField, DeviceCard | AuthController.signIn() | Mock creds (operator_rp001 / 123456 etc.); device-binding banner |
| S03 OTP | PinCodeField (6), Resend timer 30 s, Verify button | AuthController.verifyOtp() | Wrong OTP → inline error; offline → queue verify |

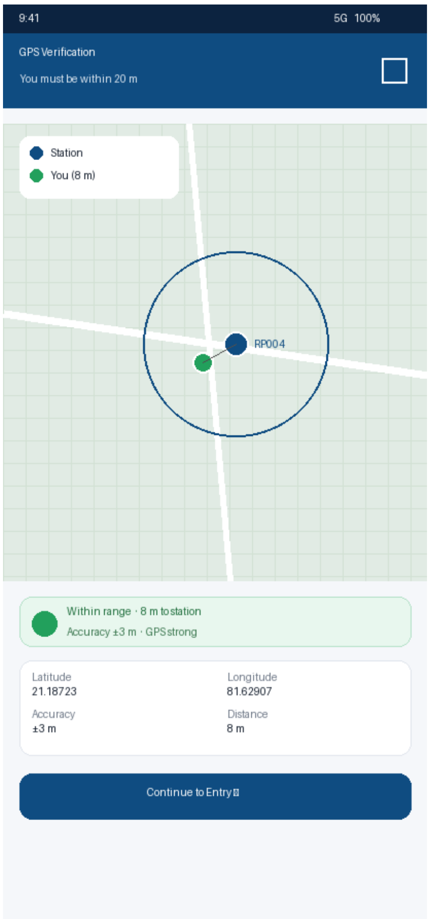
## 5.2 Field Operator



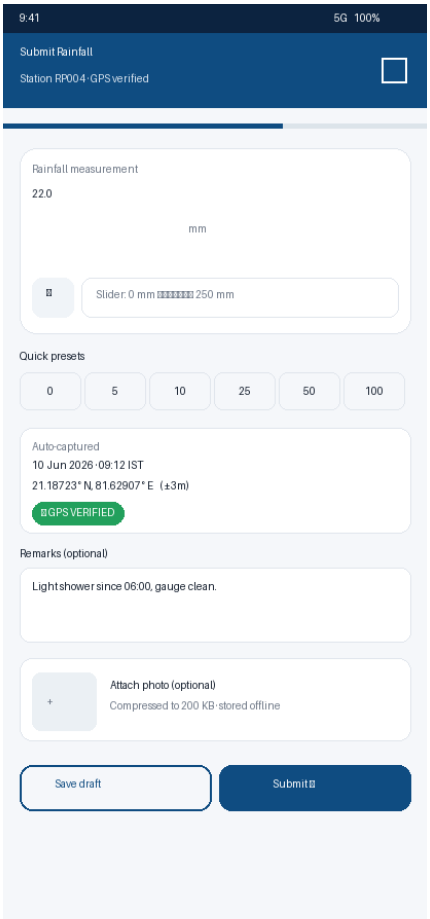

S04 — Home Dashboard



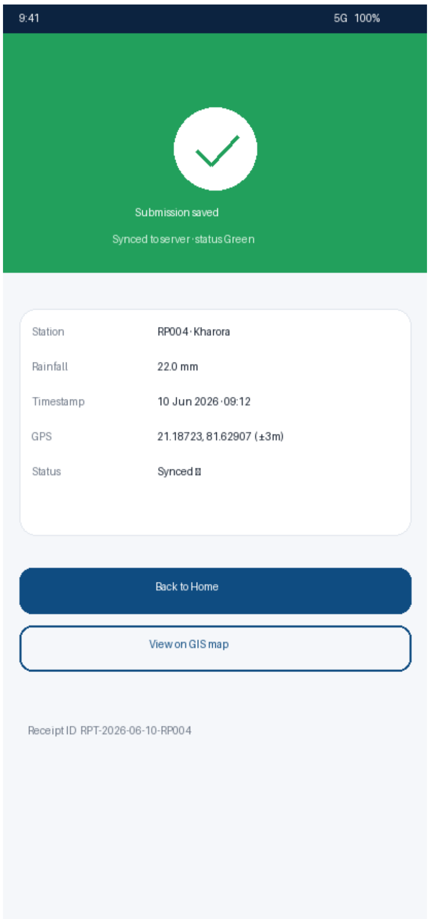

S05 — Station Detail



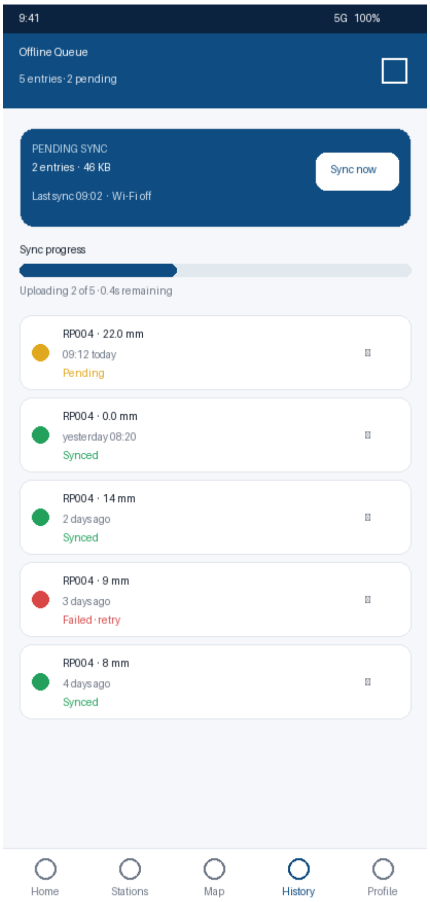

S06 — GPS Verification




S07 — Rainfall Entry



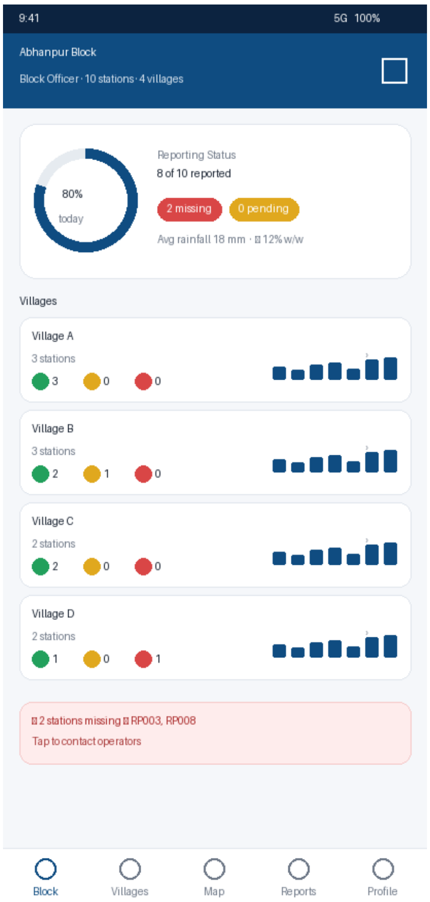

S08 — Submission Success



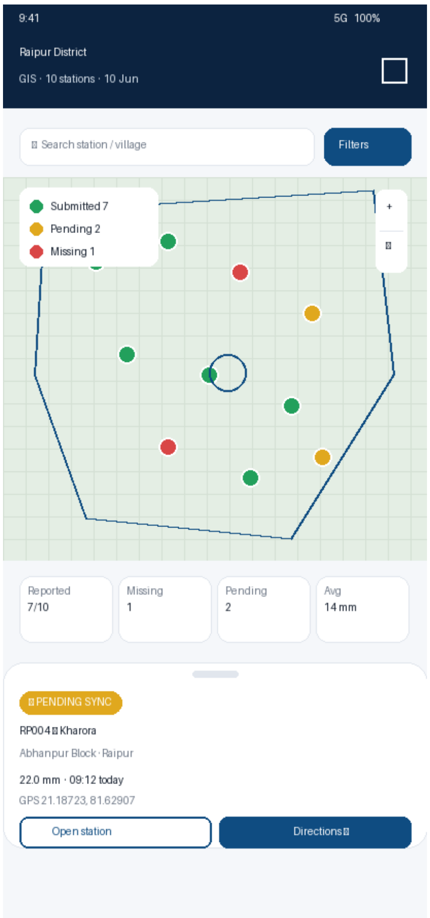

S09 — Offline Queue & Sync

| Screen | Key widgets | Data / State | Edge cases |
|---|---|---|---|
| S04 Home | StatusBanner, KpiRow ×4, MyStationCard, QuickActions, AlertCard | TodayStatusProvider, AssignedStationProvider | No submission yet, cut-off passed → red CTA |
| S05 Station Detail | StaticMiniMap, StatusChip, ReadingCard, MetaList, 7-day spark | StationProvider(id), HistoryProvider(id, 7) | Missing today → red chip; pending sync → yellow |
| S06 GPS Verify | InteractiveMap (flutter_map), AccuracyRing 20 m, DistanceBadge, ContinueCta | LocationStream, distanceBetween() | Distance > 20 m → CTA disabled, guidance toast; permission denied → settings deep-link |
| S07 Entry | BigNumberField, Slider 0–250 mm, PresetChips, AutoMetaCard, RemarksField, PhotoAttach | RainfallEntryController | Draft autosave every 5 s; on submit if offline → enqueue |
| S08 Success | SuccessHero, ReceiptList, primary+secondary CTA | Last submission state | Sync failed → receipt shows Pending banner |
| S09 Offline Queue | SyncSummaryCard, ProgressBar, QueueList(items), retry/expand | SyncQueueProvider, SyncWorker stream | Failed item → swipe-to-retry; conflict → manual resolve dialog |

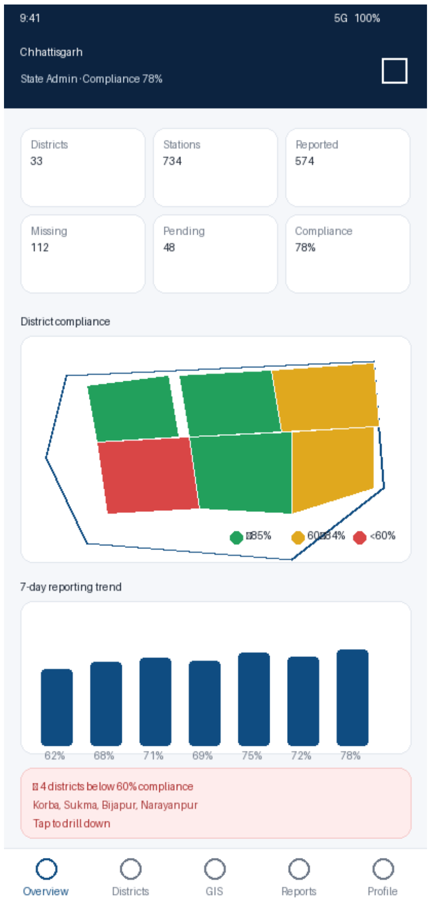
## 5.3 Block / District / State Dashboards



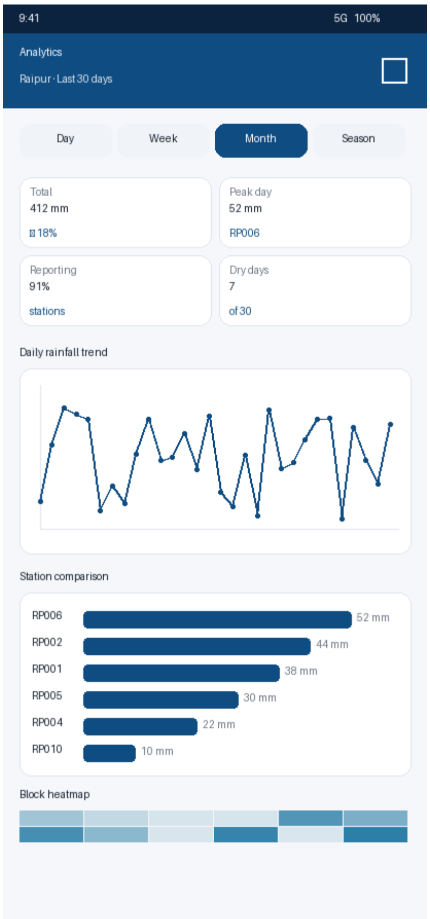

S12 — Block Officer



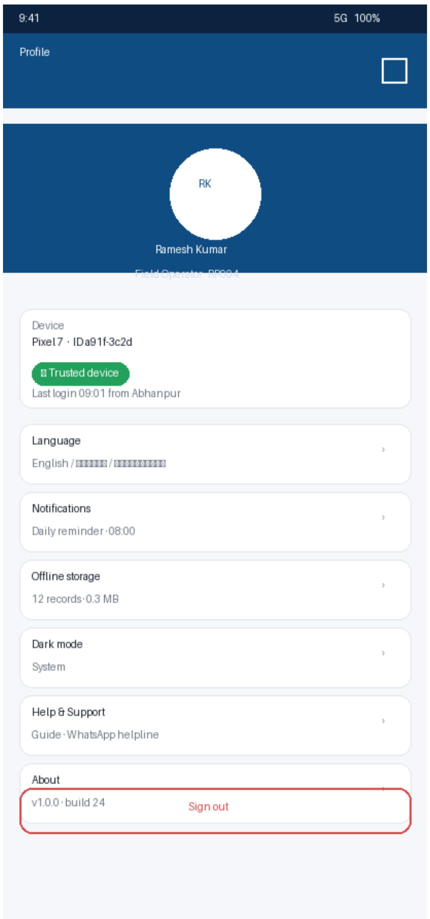

S10 — District GIS



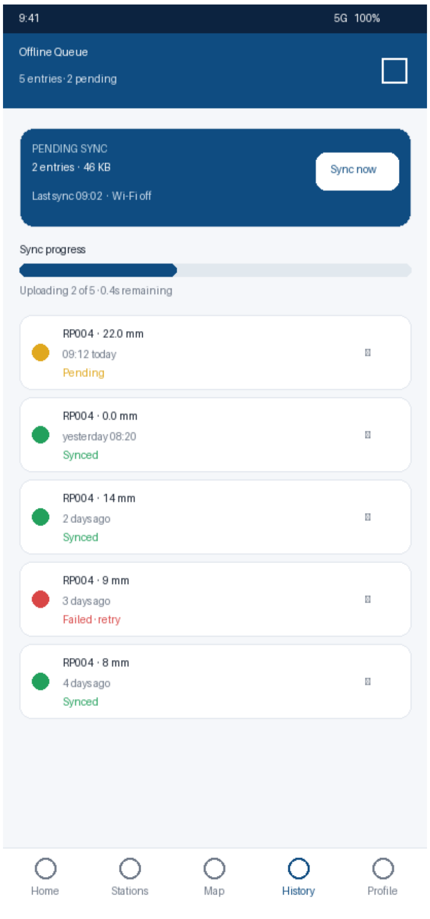

S11 — State Admin




S13 — Analytics




S14 — Profile / Settings

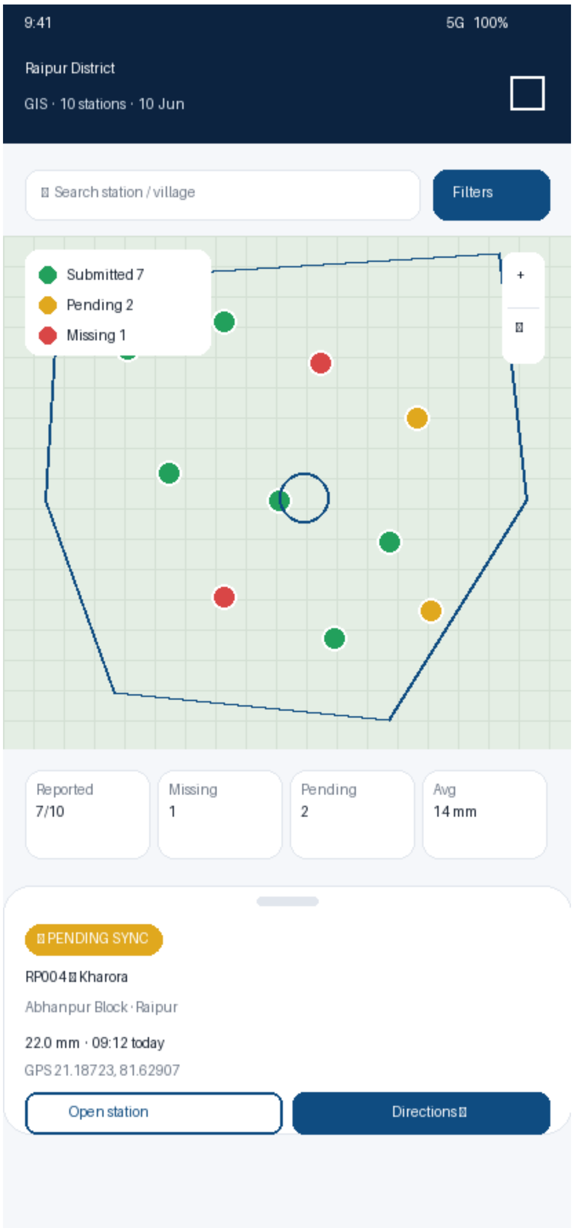



S09 — (shared) Sync Center

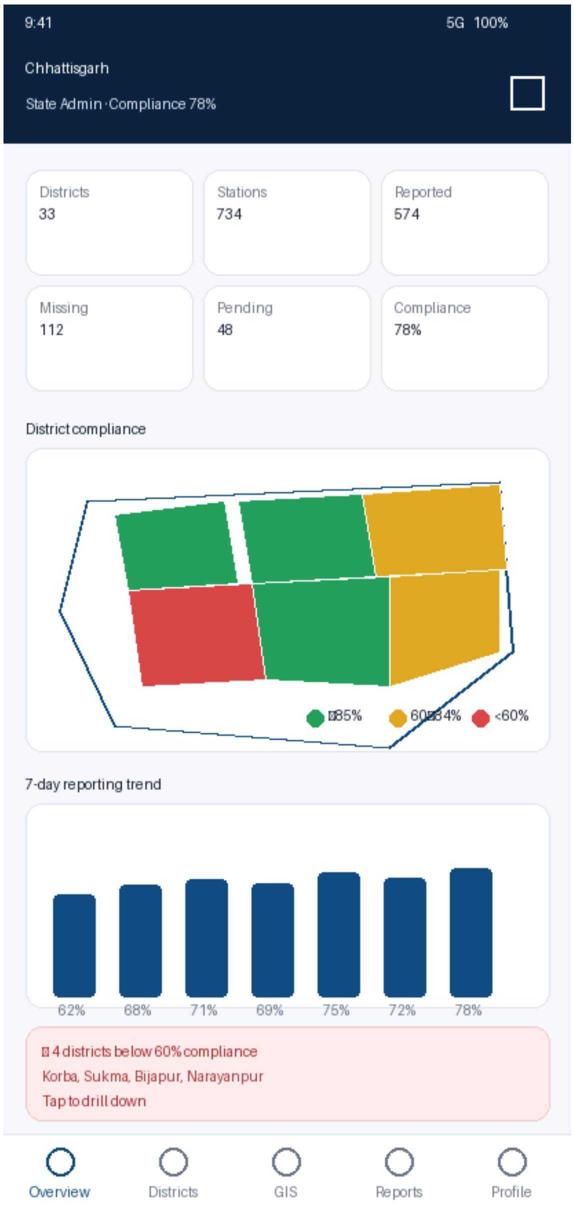
| Screen | Key widgets | Data / State | Notes |
|---|---|---|---|
| S12 Block | ComplianceRing, VillageRow ×N (with sparkline), MissingAlert | BlockMetricsProvider(blockId) | Drill-down → /block/villages/:id |
| S10 District GIS | flutter_map + MarkerClusterLayer, SearchBar, FilterSheet, KPI strip, StationBottomSheet | DistrictGisProvider | Date filter; cluster zoom; OSM tiles + cached |
| S11 State Dashboard | KpiGrid ×6, Choropleth (GeoJSON), TrendBars 7d, CriticalAlertCard | StateMetricsProvider | Tap district → /state/districts/:id |
| S13 Analytics | PeriodTabs, SummaryCards, LineChart (fl_chart), BarComparison, BlockHeatmap | AnalyticsProvider(scope, period) | Export CSV/PDF (post-MVP seam) |
| S14 Profile | Avatar, DeviceCard, SettingsList (lang, notif, dark, storage), Sign-out | ProfileProvider, SettingsRepository | Languages: en, hi, cgg |

# 6. GIS Dashboard — Primary Experience

The GIS view is the heart of the system. It must feel as fluid as Google Maps while being usable at 700+ markers. The design is identical across roles; only the data scope and clustering policy change.




District GIS




State Choropleth

## 6.1 Map stack

| Layer | Implementation | Purpose |
|---|---|---|
| Base tiles | OpenStreetMap via flutter_map TileLayer; cached with flutter_map_tile_caching | Offline-resilient basemap |
| Admin boundaries | GeoJsonLayer (state / district / block polygons) | Color-coded choropleth at state level |
| Markers | MarkerLayer + MarkerClusterLayer (clustering enabled above 50 markers in viewport) | Station status pins |
| User location | CurrentLocationLayer (accuracy circle) | GPS verification screen |

| Layer | Implementation | Purpose |
|---|---|---|
| 20 m radius | CircleLayer (radius=20, dashed) | Visual guardrail on Verify screen |

## 6.2 Marker semantics

| Color | Status | Meaning | Token |
|---|---|---|---|
| ● | Green | Submitted today | --color-status-green #22A05C |
| ● | Yellow | Pending sync | --color-status-yellow #E0A81E |
| ● | Red | Missing data | --color-status-red #D94646 |
| ■ | Hollow | Not yet due | --color-muted |

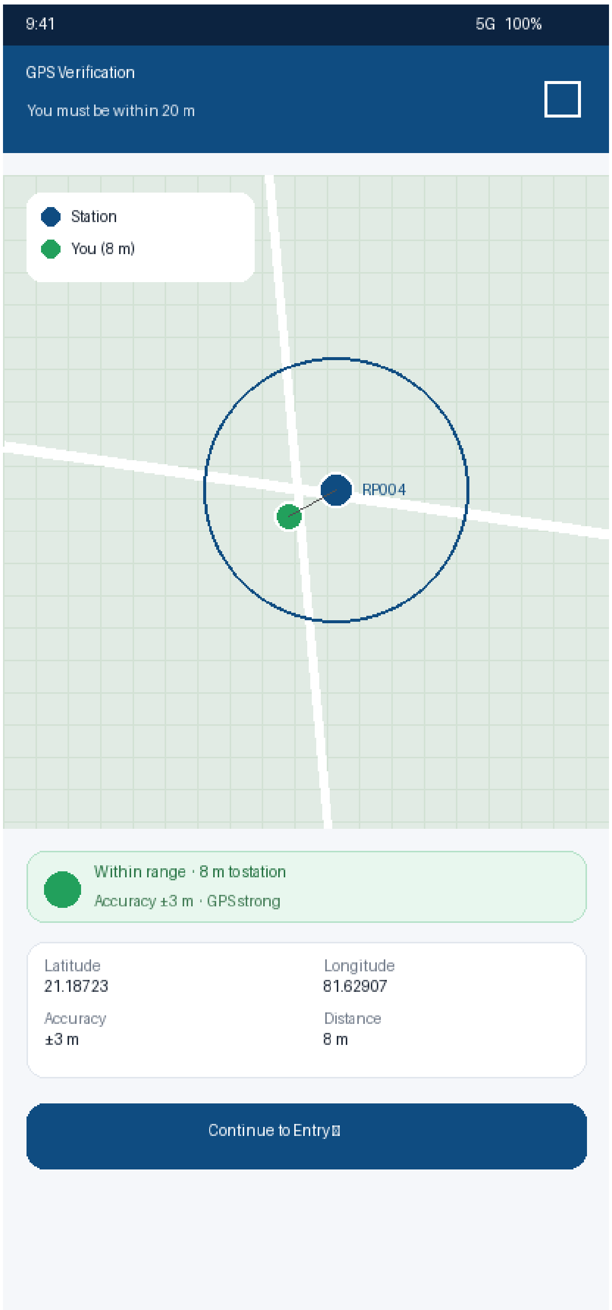
## 6.3 Station Bottom Sheet

Tapping a marker opens a Material 3 modal bottom sheet (draggable, 30%/60%/100% snap). Contents: status chip, station name, station ID, village, block, today's rainfall, last submission, GPS coordinates, primary CTA *Open station*, secondary *Directions* (handoff to Google Maps).

## 6.4 Filters

- **Date** — today (default), yesterday, custom range
- **Status** — Green / Yellow / Red toggles
- **Hierarchy** — District → Block → Village cascade
- **Search** — station ID, village name, operator name (fuzzy)

# 7. GPS Verification (≤ 20 m)




S06 — GPS Verification screen

## 7.1 Workflow

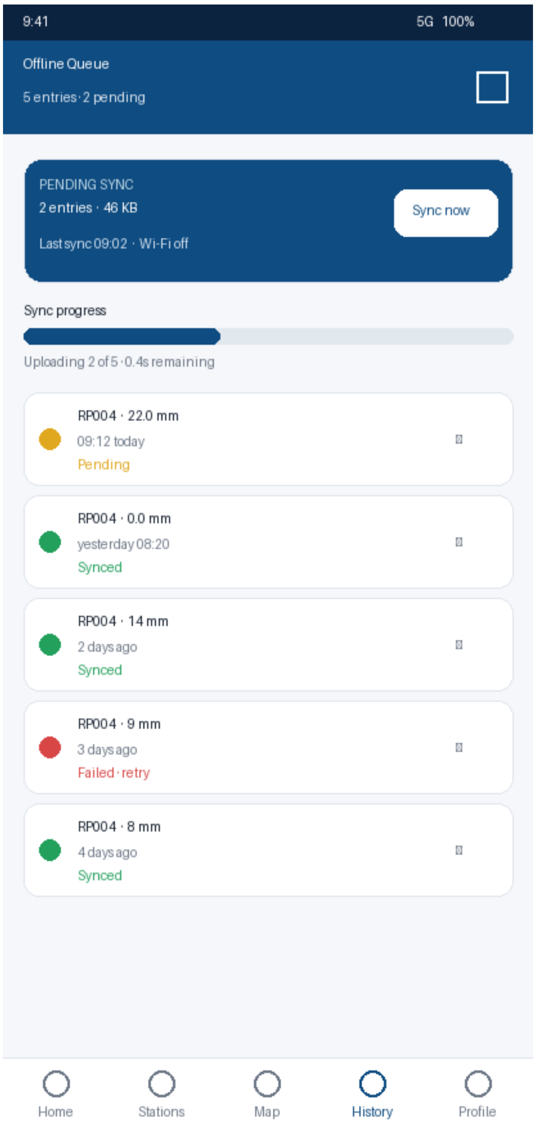
`Open station ■■ request location permission ■■ start LocationStream(5 s interval, high accuracy) ■■ compute distance = Geolocator.distanceBetween(user, station) ■■ if distance ≤ 20 m AND accuracy ≤ 15 m ■■ state=VERIFIED (green) ■■ enable Entry CTA ■■ else state=OUT_OF_RANGE (warning) ■■ disable CTA + guidance arrow`

## 7.2 Edge cases & guardrails

- Permission denied → empty state with *Open Settings* deep-link.
- Mock-location detected → block submission, log to audit (post-MVP).
- Accuracy worse than 15 m for 30 s → show *Move to open area* tip.

- Station coordinates missing → fallback to manual flag for officer review.
- Battery saver killing GPS → show *Disable battery saver* hint.

# 8. Offline-First Experience




S09 — Offline Queue & Sync Center

## 8.1 States

| State | Color | Trigger | Recovery |
|---|---|---|---|
| Synced | Green | Upload acknowledged by server | — |
| Pending | Yellow | Saved locally, no connectivity | Auto-retry on connectivity_plus stream |
| Failed | Red | HTTP 4xx/5xx after 5 retries | Manual *Retry* in queue · open conflict resolver |
| Conflict | Red (striped) | Server has newer entry for same key | Show diff, user picks *Keep mine / Use server* |

## 8.2 Sync strategy

- **Local store:** Hive box rainfall_entries, indexed by clientId (UUIDv4) + stationId + dateKey.
- **Worker:** WorkManager periodic task every 15 min + on connectivity gain.

- **Batching:** up to 50 entries per request, exponential backoff (1m → 30m).
- **Idempotency:** server keyed by (stationId, dateKey, clientId).
- **UI hooks:** top app-bar shows global sync chip (count + spinner).

# 9. Design System

A self-contained Flutter package rms_design exposes tokens, ThemeData (light + dark), components, and icons. Built on Material 3.

## 9.1 Color tokens

| Token | Light | Dark | Usage |
|---|---|---|---|
| primary | #0F4C81 | #5CBDB9 | App bars, CTAs |
| primaryContainer | #D7E7F3 | #0C2340 | Containers, chips |
| surface | #FFFFFF | #0E1620 | Cards |
| surfaceVariant | #F5F7FA | #1A2330 | Inputs, secondary cards |
| onSurface | #14202D | #E8EEF5 | Body text |
| onSurfaceMuted | #6E7887 | #8A95A4 | Captions |
| statusGreen | #22A05C | #3FD08A | Submitted |
| statusYellow | #E0A81E | #F0C24A | Pending sync |
| statusRed | #D94646 | #F46B6B | Missing / failed |
| accent | #2D8A9E | #5CBDB9 | Highlights, charts |
| outline | #E1E6ED | #2A3442 | Borders |

## 9.2 Typography

**Inter** (fallback Poppins). 1.25 type scale.

| Style | Size / Weight | Use |
|---|---|---|
| displayLarge | 32 / 700 | Splash, hero stats |
| headlineMedium | 22 / 700 | Page titles |
| titleLarge | 18 / 600 | Card headers |
| bodyLarge | 16 / 400 | Primary body |
| bodyMedium | 14 / 400 | Default body |
| labelLarge | 13 / 600 | Buttons, chips |
| caption | 11 / 400 | Meta / muted |

## 9.3 Spacing & radius

- **Grid:** 4 px base → s4, s8, s12, s16, s20, s24, s32, s40, s48.
- **Radius:** r8 (chips), **r12 (default)**, r16 (cards), r20 (sheets), full (avatar).
- **Elevation:** e0 (flat), e1 (cards), e3 (sheets), e6 (dialogs).
- **Touch target:** minimum 48 × 48 dp; primary CTA height 48–56 dp.

## 9.4 Components

| Component | Description |
|---|---|
| RmsAppBar | Color-aware (primary / primaryDark), supports subtitle + actions |
| RmsButton (primary, secondary, ghost, destructive) | 48 dp height, 12 dp radius, ripple + haptic |
| KpiCard | Label (caption) + value (titleLarge) + trend chip |
| StatusChip | Filled chip with dot + label, color from StationStatus |
| StationCard | Avatar (RPxxx), title, subtitle, status chip, trailing chevron |
| RmsTextField / PinCodeField / Slider / SegmentedTabs | Material 3 themed wrappers |
| MapMarker (station / cluster / user) | Custom painter; cluster shows count |
| StationBottomSheet | Drag handle, status chip, info grid, dual CTA |
| SyncBadge | Global app-bar chip with count + spinner |
| EmptyState / ErrorState / LoadingState | Standardised for every list/route |
| RmsChart (line, bar, heatmap) | fl_chart wrappers with token-driven colors |

## 9.5 Accessibility

- WCAG AA contrast in both themes (verified for status colors on surfaces).
- Min font scale support 0.85×–1.5× without overflow.
- Semantic labels on every map marker, button, and chip.
- Outdoor mode: optional High-Contrast theme (darker primary + thicker borders).

# 10. Flutter Architecture

## 10.1 Layered structure

```
lib/ ■■■ main.dart ■■■ app.dart (MaterialApp.router + ThemeMode) ■■■ core/ ■ ■■■ config/
(env, flavors) ■ ■■■ theme/ (rms_design tokens, light/dark) ■ ■■■ router/ (GoRouter + RoleGuard
redirect) ■ ■■■ di/ (Riverpod overrides per flavor) ■ ■■■ services/ ■ ■ ■■■
location_service.dart (Geolocator wrapper) ■ ■ ■■■ device_service.dart (device_info_plus) ■ ■
■■■ connectivity_service.dart ■ ■ ■■■ notification_service.dart (firebase_messaging —
post-MVP) ■ ■ ■■■ audit_logger.dart (post-MVP) ■ ■■■ utils/ (formatters, distance, dateKey)
■■■ data/ ■ ■■■ local/ (Hive boxes, adapters, migrations) ■ ■■■ remote/ (Dio + Retrofit
clients — mocked in MVP) ■ ■■■ repositories/ (AuthRepo, StationRepo, RainfallRepo, MetricsRepo)
■ ■■■ sync/ (SyncWorker, ConflictResolver) ■■■ domain/ ■ ■■■ models/ (Station, RainfallEntry,
User, Role…) ■ ■■■ enums/ (StationStatus, SyncStatus, Role) ■ ■■■ usecases/ (SubmitRainfall,
VerifyGps, AggregateBlock…) ■■■ features/ ■ ■■■ auth/ (login, otp, controllers) ■ ■■■
operator/ (home, station, gps, entry, queue, history) ■ ■■■ block/ (dashboard, villages, map) ■
■■■ district/ (dashboard, blocks, map, analytics) ■ ■■■ state/ (dashboard, districts, map,
analytics) ■ ■■■ gis/ (shared MapView, StationBottomSheet, filters) ■ ■■■ analytics/ (charts,
period selector) ■ ■■■ shared/ (notifications, settings, help, profile) ■■■ design/ (rms_design
— could be a /packages/rms_design package)
```

## 10.2 State management — Riverpod

| Provider | Type | Responsibility |
|---|---|---|
| authControllerProvider | StateNotifierProvider | Login, OTP, token, role, device binding |
| sessionProvider | Provider | Current user, role, assigned scope |
| assignedStationProvider | FutureProvider | Operator's single station (MVP) |
| stationProvider.family(id) | FutureProvider.family | Any station detail |
| rainfallControllerProvider.family(stationId) | StateNotifierProvider.family | Form state, submit, draft autosave |
| gpsStreamProvider | StreamProvider | Geolocator stream |
| gpsVerifyProvider.family(stationId) | Provider.family | Distance + accuracy → VERIFIED/OUT_OF_RANGE |
| syncQueueProvider | StateNotifierProvider | Pending/failed list, retry |
| connectivityProvider | StreamProvider | Drives auto-sync |
| metricsProvider.family(scope) | FutureProvider.family | Aggregations (block/district/state) |
| gisMarkersProvider.family(scope,date) | FutureProvider.family | Markers for current map viewport |
| analyticsProvider.family(scope,period) | FutureProvider.family | Charts data |
| settingsControllerProvider | StateNotifierProvider | Lang, theme, notifications |

## 10.3 Routing — GoRouter (excerpt)

```
final router = GoRouter( initialLocation: '/splash', redirect: (ctx, state) =>
RoleGuard.redirect(ctx, state), routes: [ GoRoute(path: '/splash', builder: (_, __) =>
SplashScreen()), GoRoute(path: '/auth/login', builder: ...), GoRoute(path: '/auth/otp', builder:
```

```
...), ShellRoute( /* /operator */ builder: OperatorShell, routes: [...] ), ShellRoute( /* /block
*/ builder: BlockShell, routes: [...] ), ShellRoute( /* /district */ builder: DistrictShell,
routes: [...] ), ShellRoute( /* /state */ builder: StateShell, routes: [...] ), ], );
```

**RoleGuard** redirects on every navigation: unauthenticated → /auth/login; authenticated but accessing out-of-role branch → role home; deep-link with permitted scope is allowed.

## 10.4 Key domain models

```
@HiveType(typeId: 1) class Station { final String id; // RP004 final String village; final String
block; final String district; final double lat, lng; final String operatorId; } @HiveType(typeId:
2) class RainfallEntry { final String clientId; // UUID v4 final String stationId; final double
mm; final DateTime timestamp; final double lat, lng; final double accuracyM; final String?
remarks; final String? photoPath; final SyncStatus status; // synced | pending | failed |
conflict } enum Role { fieldOperator, blockOfficer, districtOfficer, stateAdmin } enum
StationStatus { green, yellow, red, notDue }
```

## 10.5 Package pins (pubspec.yaml)

```
flutter: sdk: flutter flutter_riverpod: ^2.5.1 go_router: ^14.0.0 flutter_map: ^7.0.0
flutter_map_marker_cluster: ^1.3.6 latlong2: ^0.9.1 geolocator: ^12.0.0 permission_handler:
^11.3.1 hive_flutter: ^1.1.0 fl_chart: ^0.68.0 connectivity_plus: ^6.0.3 workmanager: ^0.5.2
device_info_plus: ^10.1.0 intl: ^0.19.0 dio: ^5.5.0 # prod
```

# 11. Component Tree (selected screens)

## 11.1 District GIS

```
DistrictGisScreen ■■■ Scaffold ■■■ RmsAppBar(title: 'Raipur District', subtitle: '10 stations ·
10 Jun', dark: true) ■■■ SearchFilterBar (RmsTextField + Filters button → FilterSheet) ■■■
Expanded ■ ■■■ Stack ■ ■■■ FlutterMap ■ ■ ■■■ TileLayer(OSM, cache) ■ ■ ■■■
GeoJsonLayer(districtBoundary) ■ ■ ■■■ MarkerClusterLayer(stationMarkers) ■ ■■■
Positioned(top, left): MapLegend(green/yellow/red) ■ ■■■ Positioned(top, right):
MapZoomControls ■■■ KpiStrip([reported, missing, pending, avg]) ■■■ StationBottomSheet
(showModalBottomSheet on marker tap)
```

## 11.2 Rainfall Entry

```
RainfallEntryScreen(stationId) ■■■ Scaffold ■■■ RmsAppBar(title: 'Submit Rainfall', subtitle:
'Station RP004 · GPS verified') ■■■ ProgressIndicator(value: 0.66) ■■■ SingleChildScrollView
■■■ BigNumberField(value: mm, formatter: 1 decimal) ■■■ Slider(0..250) +
PresetChipsRow([0,5,10,25,50,100]) ■■■ AutoMetaCard(timestamp, lat, lng, gpsVerifiedChip) ■■■
RmsTextField(label: 'Remarks', maxLines: 3) ■■■ PhotoAttach(optional, compressed to 200 KB) ■■■
ButtonRow(SaveDraft, Submit) // sticky bottom
```

# 12. Dark Mode & Tablet

## 12.1 Dark mode

Driven by ThemeMode.system with manual override in Profile. Color tokens flip via ColorScheme.dark() generated from the same source. Map switches to a dark tile style (CartoDB Dark Matter) and marker borders thicken for outdoor legibility. Status colors stay perceptually identical (slight lift in luminance).

## 12.2 Tablet (≥ 600 dp)

- **Two-pane GIS:** map fills 60% on the right; left rail shows station list with live filters.
- **Dashboards:** KPI strip switches from 2 columns to 4–6; charts grow to full width.
- **Operator entry:** form on the left, station mini-map on the right.
- **NavigationRail** replaces bottom nav above 720 dp; collapsible labels.
- **Responsive breakpoints:** compact < 600, medium 600–840, expanded > 840 (Material 3 window-size classes).

# 13. Scalability Strategy

| Concern | MVP (10 stations) | Production (734+) | Mitigation |
|---|---|---|---|
| Map markers | Plain MarkerLayer | Clustering required | MarkerClusterLayer with viewport-bound query |
| List rendering | ListView | Long lists | slivers + pagination (cursor-based) |
| State aggregation | On-device | Server-side | AggregationApi precomputed nightly + delta cache |
| Offline payload | Hive box | Per-block partition | Open one box per assigned scope; LRU eviction |
| Auth | Mock | OAuth2 / PKCE + device binding | AuthRepository swap; no UI change |
| Notifications | None | Push (FCM), missing-station alerts | NotificationService seam (already in tree) |
| Realtime | Polling | WebSocket / Supabase Realtime | RealtimeChannel + Riverpod stream provider |
| Tile cache | Memory only | Disk + tile pre-bundling | flutter_map_tile_caching with bbox per district |
| Analytics | On-device sums | Heatmaps, time series | Server-side rollups; client renders only |
| Localisation | en + hi | + Chhattisgarhi, regional dialects | ARB files, RTL ready |
| Roles | 4 roles | + Auditor, IT Admin | Role enum + capability matrix (data-driven) |
| Users | Tens | Thousands | Stateless API, JWT, paginated everything |

# 14. Developer Handoff Specifications

## 14.1 Engineering standards

- **Code:** Dart 3.4, very_good_analysis lints, 100% null-safety.
- **Testing:** unit (use cases), widget (each screen happy + edge), integration (operator + officer flows). Target coverage 70%.
- **CI:** GitHub Actions → analyze, test, build apk/aab/ios.
- **Flavors:** mock (in-memory), dev, staging, prod (env-driven base URLs).
- **Crash + analytics:** Sentry + a thin AnalyticsLogger (no PII).
- **Security:** tokens in flutter_secure_storage; Hive box encrypted with HMAC-SHA256 key in keystore.

## 14.2 Acceptance criteria (MVP)

| # | Criterion |
|---|---|
| AC1 | Field operator can sign in with mock creds and reach Home in ≤ 3 s on a mid-range Android. |
| AC2 | GPS verification correctly blocks submission when distance > 20 m (tested with mocked locations). |
| AC3 | Submitting offline saves to Hive within 200 ms and shows Pending chip. |
| AC4 | On reconnect, sync uploads and station marker turns Green within 5 s on the GIS. |
| AC5 | Block officer dashboard shows correct counts (reported/missing/pending) for seeded data. |
| AC6 | District GIS renders 10 markers with status colors and opens bottom sheet on tap. |
| AC7 | State dashboard renders choropleth + 7-day trend; tapping a district drills down. |
| AC8 | All screens pass accessibility audit at AA contrast and 1.3× text scale. |
| AC9 | No crash under airplane-mode end-to-end operator flow. |
| AC10 | App builds for Android 8+ and iOS 14+ with no analyzer warnings. |

## 14.3 Deliverables

- This specification (PDF) — source of truth.
- rms_design Flutter package — tokens, theme, components.
- Annotated Figma equivalent (mockups in §5 are the canonical reference for the MVP).
- Seed data: 10 stations, 4 villages, 3 blocks (Raipur), 4 mock users.
- Postman collection for the future API (request/response stubs match repositories).

## 14.4 Out of scope (MVP)

- Real authentication backend (mocked).
- Push notifications, SMS escalations.
- Heatmaps, advanced analytics dashboards.
- Audit logs, role management UI.
- Multi-tenant deployment beyond Raipur.

## 14.5 Glossary

| Term | Meaning |
|---|---|
| Station | Physical rainfall gauge identified by code (e.g. RP004) |
| Dry day | Day with rainfall < 0.1 mm |
| Cut-off | Daily reporting deadline (08:30 IST) |
| Compliance | % stations that reported on a given day |
| dateKey | UTC date key (YYYY-MM-DD) used to dedupe entries |
| clientId | UUID v4 generated on device for idempotent sync |

# Appendix A — Screen Index

| ID | Screen | Roles | Section |
|---|---|---|---|
| S01 | Splash | All | §5.1 |
| S02 | Login (Mobile+OTP / Username) | All | §5.1 |
| S03 | OTP Verification | All (when OTP) | §5.1 |
| S04 | Operator Home | Field Operator | §5.2 |
| S05 | Station Detail | Field Operator | §5.2 |
| S06 | GPS Verification | Field Operator | §5.2 / §7 |
| S07 | Rainfall Entry | Field Operator | §5.2 |
| S08 | Submission Success | Field Operator | §5.2 |
| S09 | Offline Queue / Sync Center | Field Operator | §5.2 / §8 |
| S10 | District GIS Dashboard | District / State | §5.3 / §6 |
| S11 | State Dashboard | State Admin | §5.3 |
| S12 | Block Officer Dashboard | Block Officer | §5.3 |
| S13 | Analytics | Block / District / State | §5.3 |
| S14 | Profile / Settings | All | §5.3 |

— End of specification —